To estimate the **yearly profit** of a **500 MW/year sand-to-solar-cell integrated polycrystalline silicon plant**, we need to calculate:

Revenue − Operating Cost − Depreciation/Finance/Tax

I will use a realistic 2026 Chinese PV industry scenario.

Assumptions

Production:

- Solar cell output: **500 MW/year**
- Operating rate: 90%
- Actual saleable output: **450 MW/year**
- Technology: Polycrystalline silicon cell
- Plant scope: **Quartz sand → polysilicon → wafer → solar cell**
- No module assembly included

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1. Revenue

Solar cell selling price (not module):

Current market range:

$0.08–0.12/W

Assume:

[
0.10 USD/W
]

Annual sales:

[
450,000,000W \times 0.10
]

=

$45 million/year

---

2. Production Cost

From previous calculation:

Sand-to-cell manufacturing cost:

≈ **$0.25–0.35/W**

Assume:

[
0.30 USD/W
]

Annual cost:

[
450MW \times 1,000,000 \times 0.30
]

=

$135 million/year

---

3. Operating Result

Revenue:

$45M

minus production cost:

$135M

=

-$90 million/year loss

Why?

Because **500 MW integrated polysilicon production is too small**.

The expensive steps are:

- Polysilicon purification
- Chemical recycling
- Siemens reactors
- Hydrogen/chlorosilane systems

They require GW-scale factories.

If Solar Cell Price Recovers

Assume selling price:

$0.18/W

Revenue:

[
450M \times 0.18
]

= $81M

Cost:

$135M

Result:

-$54M/year

Still loss.

Break-even Analysis

For a 500 MW integrated plant:

Required selling price:

$$
\frac{135M}{450M W}
$$

=

$0.30/W

Current solar cell prices are far below this.

Realistic Profit Scenario

A profitable integrated plant needs much larger scale.

Example:

10 GW/year integrated PV plant

Approximate:

Item	Value
Annual production	10 GW
CAPEX	$4–6 billion
Manufacturing cost	$0.08–0.12/W
Selling price	$0.12–0.15/W

Annual revenue:

[
10,000MW \times 0.13
]

=

$1.3 billion/year

Cost:

[
10,000MW \times 0.10
]

=

$1 billion/year

Operating profit:

$200–300 million/year

For Your 500 MW Project

A realistic financial conclusion:

Item	Value
Investment	$400–450M
Annual sales	$45–60M
Annual operating cost	$120–150M
Profit	□ Loss
Payback	Impossible

Better business model

For a 500 MW factory, I would not recommend starting from sand.

More profitable configuration:

Option A:

Buy polysilicon → wafer → cell

CAPEX:
≈ $100M

Possible profit:
≈ **$10–25M/year**

Option B:

Buy wafers → make cells

CAPEX:
≈ $50–70M

Possible profit:
≈ **$5–15M/year**

Option C:

Sand-to-polysilicon only at >20,000 ton/year

Then integrate downstream.

For a serious investment decision, the next calculation should be a **10-year financial model (CAPEX, depreciation, loan interest, IRR, NPV, payback period)** because the $400M investment changes the economics completely.